

IGNOU BCS-012: Basic Mathematics**Term-End Examination, December 2025****Full Solved Paper with Theory/Formula Sheet**

Time: 3 Hours Maximum Marks: 100

Note

Question No. 1 is compulsory. Attempt any three questions from the remaining questions. For practice, solutions to all questions are provided.

Question 1**Question**

(a) Show that

$$\begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ca & cb & -c^2 \end{vmatrix} = 4a^2b^2c^2.$$

Solution. Take factors a, b, c from the first, second and third rows respectively:

$$\Delta = abc \begin{vmatrix} -a & b & c \\ a & -b & c \\ a & b & -c \end{vmatrix}.$$

Expanding the smaller determinant along the first row,

$$\begin{aligned} D &= (-a)\{(-b)(-c) - bc\} - b\{a(-c) - ac\} + c\{ab - (-b)a\} \\ &= (-a)(bc - bc) - b(-2ac) + c(2ab) = 0 + 2abc + 2abc = 4abc. \end{aligned}$$

Therefore,

$$\Delta = abc(4abc) = 4a^2b^2c^2.$$

$$\boxed{\Delta = 4a^2b^2c^2}$$

Question

(b) If

$$A = \begin{pmatrix} -1 & -2 \\ 3 & 6 \end{pmatrix},$$

show that $A^2 = 5A$.

Solution.

$$A^2 = \begin{pmatrix} -1 & -2 \\ 3 & 6 \end{pmatrix} \begin{pmatrix} -1 & -2 \\ 3 & 6 \end{pmatrix} = \begin{pmatrix} 1-6 & 2-12 \\ -3+18 & -6+36 \end{pmatrix} = \begin{pmatrix} -5 & -10 \\ 15 & 30 \end{pmatrix}.$$

But

$$5A = 5 \begin{pmatrix} -1 & -2 \\ 3 & 6 \end{pmatrix} = \begin{pmatrix} -5 & -10 \\ 15 & 30 \end{pmatrix}.$$

Hence $A^2 = 5A$.

Question

(c) Use the principle of mathematical induction to show that

$$1 + 3 + \cdots + (2n - 1) = n^2, \quad \forall n \in \mathbb{N}.$$

Solution. Let $P(n)$ be the statement

$$1 + 3 + \cdots + (2n - 1) = n^2.$$

For $n = 1$, LHS = 1 and RHS = $1^2 = 1$. Hence $P(1)$ is true.

Assume $P(k)$ is true, that is,

$$1 + 3 + \cdots + (2k - 1) = k^2.$$

For $n = k + 1$,

$$\begin{aligned} 1 + 3 + \cdots + (2k - 1) + \{2(k + 1) - 1\} &= k^2 + (2k + 1) \\ &= k^2 + 2k + 1 = (k + 1)^2. \end{aligned}$$

Thus $P(k + 1)$ is true whenever $P(k)$ is true. Hence, by induction,

$$1 + 3 + \cdots + (2n - 1) = n^2 \quad \forall n \in \mathbb{N}.$$

Question

(d) If $m \neq n$ and m times the m th term of an A.P. is equal to n times the n th term of the A.P., show that the $(m + n)$ th term is zero.

Solution. Let the first term be a and common difference be d . Then

$$T_r = a + (r - 1)d.$$

Given

$$mT_m = nT_n.$$

So

$$m\{a + (m - 1)d\} = n\{a + (n - 1)d\}.$$

Expanding,

$$ma + m(m - 1)d = na + n(n - 1)d.$$

Bring all terms to one side:

$$(m - n)a + \{m(m - 1) - n(n - 1)\}d = 0.$$

Now,

$$m(m - 1) - n(n - 1) = m^2 - m - n^2 + n = (m - n)(m + n - 1).$$

Therefore,

$$(m - n)\{a + (m + n - 1)d\} = 0.$$

Since $m \neq n$, $m - n \neq 0$. Hence

$$a + (m + n - 1)d = 0.$$

But

$$T_{m+n} = a + (m + n - 1)d.$$

Therefore,

$$\boxed{T_{m+n} = 0}.$$

Question

(e) If $z \in \mathbb{C}$ and $|z - i| = |z + i|$, show that $\text{Im}(z) = 0$.

Solution. Let

$$z = x + iy, \quad x, y \in \mathbb{R}.$$

Then

$$z - i = x + i(y - 1), \quad z + i = x + i(y + 1).$$

Using the modulus formula,

$$|z - i|^2 = x^2 + (y - 1)^2, \quad |z + i|^2 = x^2 + (y + 1)^2.$$

Given $|z - i| = |z + i|$, so their squares are equal:

$$x^2 + (y - 1)^2 = x^2 + (y + 1)^2.$$

Cancelling x^2 ,

$$y^2 - 2y + 1 = y^2 + 2y + 1.$$

Therefore

$$-2y = 2y \Rightarrow 4y = 0 \Rightarrow y = 0.$$

Since $\text{Im}(z) = y$, we get

$$\boxed{\text{Im}(z) = 0}.$$

Question

(f) Find the roots of

$$x^3 - 13x^2 + 15x + 189 = 0,$$

given that one root exceeds the other by 2 and the roots are integers.

Solution. By trial among integer factors of 189, put $x = 7$:

$$7^3 - 13(7^2) + 15(7) + 189 = 343 - 637 + 105 + 189 = 0.$$

So $x = 7$ is a root and $(x - 7)$ is a factor. Dividing,

$$x^3 - 13x^2 + 15x + 189 = (x - 7)(x^2 - 6x - 27).$$

Now

$$x^2 - 6x - 27 = 0 \Rightarrow x = \frac{6 \pm \sqrt{36 + 108}}{2} = \frac{6 \pm 12}{2}.$$

Thus

$$x = 9 \quad \text{or} \quad x = -3.$$

The roots are

$$\boxed{7, 9, -3}.$$

Here 9 exceeds 7 by 2, as required.

Question

(g) Evaluate

$$I = \int \frac{x^3}{(x+1)^2} dx.$$

Solution. First divide x^3 by $(x+1)^2 = x^2 + 2x + 1$:

$$\frac{x^3}{(x+1)^2} = x - 2 + \frac{3x+2}{(x+1)^2}.$$

Write $3x+2 = 3(x+1) - 1$. Hence

$$\frac{3x+2}{(x+1)^2} = \frac{3}{x+1} - \frac{1}{(x+1)^2}.$$

Therefore

$$\begin{aligned} I &= \int \left(x - 2 + \frac{3}{x+1} - \frac{1}{(x+1)^2} \right) dx \\ &= \frac{x^2}{2} - 2x + 3 \ln |x+1| + \frac{1}{x+1} + C. \end{aligned}$$

$$\boxed{I = \frac{x^2}{2} - 2x + 3 \ln |x+1| + \frac{1}{x+1} + C}$$

Question

(h) Show that the diagonals of a rhombus are at right angles.

Solution. Let a rhombus have adjacent side vectors $\vec{a}$ and $\vec{b}$. Since all sides of a rhombus are equal,

$$|\vec{a}| = |\vec{b}|.$$

The two diagonals are represented by

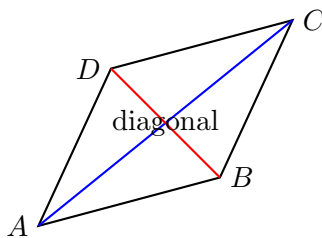
$$\vec{a} + \vec{b} \quad \text{and} \quad \vec{a} - \vec{b}.$$

Their dot product is

$$(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = \vec{a} \cdot \vec{a} - \vec{b} \cdot \vec{b} = |\vec{a}|^2 - |\vec{b}|^2 = 0.$$

Since the dot product is zero, the diagonals are perpendicular.

Diagonals of a rhombus are at right angles.



Question 2

Question

(a) Find the 10th term of the Harmonic Progression

$$\frac{1}{5}, \frac{1}{11}, \frac{1}{17}, \frac{1}{23}, \dots$$

Solution. In a harmonic progression, reciprocals of the terms form an arithmetic progression. The reciprocals are

$$5, 11, 17, 23, \dots$$

This is an A.P. with first term 5 and common difference 6. The 10th term of this A.P. is

$$A_{10} = 5 + (10 - 1)6 = 5 + 54 = 59.$$

Therefore the 10th term of the given H.P. is

$$\boxed{\frac{1}{59}}.$$

Question

(b) If $x = a + b$, $y = a\omega + b\omega^2$ and $z = a\omega^2 + b\omega$, show that $xyz = a^3 + b^3$, where $\omega \neq 1$ is a cube root of unity.

Solution. For cube roots of unity,

$$1 + \omega + \omega^2 = 0, \quad \omega^3 = 1.$$

Now

$$xyz = (a + b)(a\omega + b\omega^2)(a\omega^2 + b\omega).$$

First multiply the last two factors:

$$\begin{aligned} (a\omega + b\omega^2)(a\omega^2 + b\omega) &= a^2\omega^3 + ab\omega^2 + ab\omega^4 + b^2\omega^3 \\ &= a^2 + ab(\omega^2 + \omega) + b^2 \\ &= a^2 - ab + b^2. \end{aligned}$$

Therefore,

$$xyz = (a + b)(a^2 - ab + b^2) = a^3 + b^3.$$

Hence proved.

Question

(c) Find the direction cosines of the line joining $(0, 1, -1)$ and $(3, 2, 1)$.

Solution. Direction ratios of the line are obtained by subtracting coordinates:

$$(3 - 0, 2 - 1, 1 - (-1)) = (3, 1, 2).$$

The magnitude is

$$\sqrt{3^2 + 1^2 + 2^2} = \sqrt{14}.$$

Therefore the direction cosines are

$$\boxed{\left(\frac{3}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}\right)}.$$

Question

(d) If

$$y = \frac{e^x + e^{-x}}{e^x - e^{-x}},$$

find $\frac{dy}{dx}$.

Solution. Let

$$u = e^x + e^{-x}, \quad v = e^x - e^{-x}.$$

Then

$$u' = e^x - e^{-x} = v, \quad v' = e^x + e^{-x} = u.$$

Using the quotient rule,

$$\frac{dy}{dx} = \frac{vu' - uv'}{v^2} = \frac{v^2 - u^2}{v^2}.$$

Now

$$v^2 - u^2 = (v - u)(v + u) = (-2e^{-x})(2e^x) = -4.$$

Hence

$$\boxed{\frac{dy}{dx} = \frac{-4}{(e^x - e^{-x})^2}}.$$

Question 3

Question

(a) If

$$y = \ln \left[e^x \left(\frac{x-2}{x+2} \right)^{3/4} \right],$$

show that

$$\frac{dy}{dx} = \frac{x^2 - 1}{x^2 - 4}.$$

Solution. Using logarithmic laws,

$$y = x + \frac{3}{4} \ln \left(\frac{x-2}{x+2} \right) = x + \frac{3}{4} \{ \ln(x-2) - \ln(x+2) \}.$$

Differentiating,

$$\frac{dy}{dx} = 1 + \frac{3}{4} \left(\frac{1}{x-2} - \frac{1}{x+2} \right).$$

Now

$$\frac{1}{x-2} - \frac{1}{x+2} = \frac{(x+2) - (x-2)}{(x-2)(x+2)} = \frac{4}{x^2 - 4}.$$

Therefore

$$\frac{dy}{dx} = 1 + \frac{3}{4} \cdot \frac{4}{x^2 - 4} = 1 + \frac{3}{x^2 - 4} = \frac{x^2 - 4 + 3}{x^2 - 4} = \frac{x^2 - 1}{x^2 - 4}.$$

Hence proved.

Question

(b) Find the intervals in which

$$f(x) = 3x^{5/2} - 5x^{3/2}, \quad x > 0,$$

is increasing and decreasing.

Solution. Differentiate:

$$f'(x) = 3 \cdot \frac{5}{2} x^{3/2} - 5 \cdot \frac{3}{2} x^{1/2} = \frac{15}{2} x^{1/2} (x - 1).$$

Since $x > 0$, $x^{1/2} > 0$. Therefore the sign of $f'(x)$ depends on $(x - 1)$.

Interval	$f'(x)$	Nature
$(0, 1)$	$-$	decreasing
$(1, \infty)$	$+$	increasing

Hence

decreasing on $(0, 1)$, increasing on $(1, \infty)$.

Question

(c) Evaluate

$$\int \frac{3^x + 2^x}{6^x} dx.$$

Solution. Since $6^x = (2 \cdot 3)^x = 2^x 3^x$,

$$\frac{3^x + 2^x}{6^x} = \frac{3^x}{2^x 3^x} + \frac{2^x}{2^x 3^x} = 2^{-x} + 3^{-x}.$$

Therefore

$$\begin{aligned} \int \frac{3^x + 2^x}{6^x} dx &= \int (2^{-x} + 3^{-x}) dx \\ &= -\frac{2^{-x}}{\ln 2} - \frac{3^{-x}}{\ln 3} + C. \end{aligned}$$

$$-\frac{2^{-x}}{\ln 2} - \frac{3^{-x}}{\ln 3} + C$$

Question

(d) If

$$\vec{a} = 4\hat{i} + \hat{j} + 3\hat{k}, \quad \vec{b} = -2\hat{i} + \hat{j} - 2\hat{k},$$

find a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$.

Solution. A vector perpendicular to both $\vec{a}$ and $\vec{b}$ is $\vec{a} \times \vec{b}$:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 1 & 3 \\ -2 & 1 & -2 \end{vmatrix}.$$

Thus

$$\vec{a} \times \vec{b} = \hat{i}(1 \cdot (-2) - 3 \cdot 1) - \hat{j}(4 \cdot (-2) - 3 \cdot (-2)) + \hat{k}(4 \cdot 1 - 1 \cdot (-2)).$$

So

$$\vec{a} \times \vec{b} = -5\hat{i} + 2\hat{j} + 6\hat{k}.$$

Its magnitude is

$$\sqrt{(-5)^2 + 2^2 + 6^2} = \sqrt{65}.$$

Therefore a required unit vector is

$$\frac{-5\hat{i} + 2\hat{j} + 6\hat{k}}{\sqrt{65}}.$$

The opposite vector is also correct.

Question 4

Question

(a) If $x \in \mathbb{R}$, solve the inequality

$$\frac{9}{x-3} < 5.$$

Solution. Bring all terms to one side:

$$\frac{9}{x-3} - 5 < 0.$$

Taking common denominator,

$$\frac{9 - 5(x-3)}{x-3} < 0 \Rightarrow \frac{24 - 5x}{x-3} < 0.$$

Critical values are

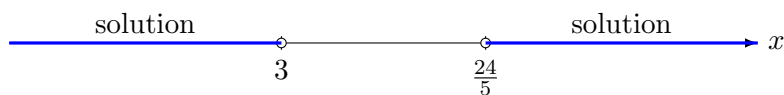
$$x = 3 \quad \text{and} \quad x = \frac{24}{5}.$$

Check signs on intervals:

Interval	$\frac{24-5x}{x-3}$
$(-\infty, 3)$	—
$(3, 24/5)$	+
$(24/5, \infty)$	—

Hence the solution set is

$$\boxed{(-\infty, 3) \cup \left(\frac{24}{5}, \infty\right)}.$$



Question

(b) If $a, b, x, y \in \mathbb{R}$, $a \neq 0$, $b \neq 0$ and

$$(a - ib)(x + iy) = (a^2 + b^2)i,$$

find x and y .

Solution. Expand the left side:

$$(a - ib)(x + iy) = ax + iay - ibx + by = (ax + by) + i(ay - bx).$$

Equating real and imaginary parts with $0 + i(a^2 + b^2)$, we get

$$ax + by = 0, \quad ay - bx = a^2 + b^2.$$

From $ax + by = 0$,

$$by = -ax.$$

The values $x = -b$ and $y = a$ satisfy

$$a(-b) + b(a) = 0,$$

and

$$a(a) - b(-b) = a^2 + b^2.$$

Therefore

$$\boxed{x = -b, \quad y = a}.$$

Question

(c) Use Cramer's rule to solve

$$2x - y + z = 4, \quad 3x - y = 5, \quad 2y - z = 1.$$

Solution. Write the system as $AX = B$:

$$A = \begin{pmatrix} 2 & -1 & 1 \\ 3 & -1 & 0 \\ 0 & 2 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad B = \begin{pmatrix} 4 \\ 5 \\ 1 \end{pmatrix}.$$

The determinant of A is

$$\Delta = \begin{vmatrix} 2 & -1 & 1 \\ 3 & -1 & 0 \\ 0 & 2 & -1 \end{vmatrix} = 1.$$

Replacing columns by B gives

$$\Delta_x = \begin{vmatrix} 4 & -1 & 1 \\ 5 & -1 & 0 \\ 1 & 2 & -1 \end{vmatrix} = 2,$$

$$\Delta_y = \begin{vmatrix} 2 & 4 & 1 \\ 3 & 5 & 0 \\ 0 & 1 & -1 \end{vmatrix} = 1,$$

$$\Delta_z = \begin{vmatrix} 2 & -1 & 4 \\ 3 & -1 & 5 \\ 0 & 2 & 1 \end{vmatrix} = 1.$$

By Cramer's rule,

$$x = \frac{\Delta_x}{\Delta} = 2, \quad y = \frac{\Delta_y}{\Delta} = 1, \quad z = \frac{\Delta_z}{\Delta} = 1.$$

$$\boxed{x = 2, \quad y = 1, \quad z = 1}$$

Question

(d) If

$$A = \begin{pmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{pmatrix},$$

show that $AB = BA = 6I_3$. Find A^{-1} .

Solution. Compute AB :

$$AB = \begin{pmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{pmatrix} = 6I_3.$$

Similarly,

$$BA = \begin{pmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{pmatrix} = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{pmatrix} = 6I_3.$$

Thus

$$AB = BA = 6I_3.$$

Dividing by 6,

$$A\left(\frac{1}{6}B\right) = I_3.$$

Therefore

$$A^{-1} = \frac{1}{6}B.$$

Hence

$$A^{-1} = \frac{1}{6} \begin{pmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{pmatrix}.$$

Question 5

Question

(a) Use mathematical induction to show that

$$1 + \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^n} < 2, \quad \forall n \in \mathbb{N}.$$

Solution. Let

$$P(n) : 1 + \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^n} < 2.$$

For $n = 1$,

$$1 + \frac{1}{2} = \frac{3}{2} < 2,$$

so $P(1)$ is true.

Assume $P(k)$ is true, i.e.

$$1 + \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^k} < 2.$$

For $n = k + 1$,

$$S_{k+1} = S_k + \frac{1}{2^{k+1}}.$$

To get a sharper bound, use the finite G.P. sum for S_k :

$$S_k = 1 + \frac{1}{2} + \cdots + \frac{1}{2^k} = 2 - \frac{1}{2^k}.$$

Thus

$$S_{k+1} = 2 - \frac{1}{2^k} + \frac{1}{2^{k+1}} = 2 - \frac{1}{2^{k+1}} < 2.$$

Therefore $P(k + 1)$ is true. Hence, by mathematical induction,

$$1 + \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^n} < 2 \quad \forall n \in \mathbb{N}.$$

Question

(b) Find the quadratic equation with real coefficients if one of its roots is $-2 + 3i$.

Solution. For a quadratic equation with real coefficients, complex roots occur in conjugate pairs. Hence the other root is

$$-2 - 3i.$$

The required equation is

$$(x - (-2 + 3i))(x - (-2 - 3i)) = 0.$$

Thus

$$(x + 2 - 3i)(x + 2 + 3i) = 0.$$

Using $(p - q)(p + q) = p^2 - q^2$,

$$(x + 2)^2 - (3i)^2 = 0.$$

Since $i^2 = -1$,

$$(x + 2)^2 + 9 = 0.$$

Therefore

$$x^2 + 4x + 4 + 9 = 0.$$

$$\boxed{x^2 + 4x + 13 = 0}$$

Question

(c) If

$$\vec{a} = -\hat{i} + \hat{j} + 2\hat{k}, \quad \vec{b} = 2\hat{i} - \hat{j} + \hat{k}, \quad \vec{c} = 3\hat{i} - \hat{j} + 2\hat{k},$$

find $(\vec{a} \times \vec{b}) \cdot \vec{c}$.

Solution. First compute

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{vmatrix}.$$

Thus

$$\vec{a} \times \vec{b} = \hat{i}(1 \cdot 1 - 2(-1)) - \hat{j}((-1) \cdot 1 - 2 \cdot 2) + \hat{k}((-1)(-1) - 1 \cdot 2).$$

So

$$\vec{a} \times \vec{b} = 3\hat{i} + 5\hat{j} - \hat{k}.$$

Now

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = (3, 5, -1) \cdot (3, -1, 2).$$

Therefore

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = 9 - 5 - 2 = 2.$$

$$\boxed{(\vec{a} \times \vec{b}) \cdot \vec{c} = 2}$$

Question

(d) Maximize

$$P = 5x + 2y$$

subject to

$$10x + 2y \geq 2100, \quad x + \frac{1}{2}y \leq 600, \quad y \leq 800, \quad x \geq 0, \quad y \geq 0.$$

Solution. Simplify the first two inequalities:

$$10x + 2y \geq 2100 \Rightarrow 5x + y \geq 1050,$$

$$x + \frac{1}{2}y \leq 600 \Rightarrow 2x + y \leq 1200.$$

The feasible region is bounded by

$$5x + y = 1050, \quad 2x + y = 1200, \quad y = 800, \quad x = 0, \quad y = 0.$$

The relevant corner points are

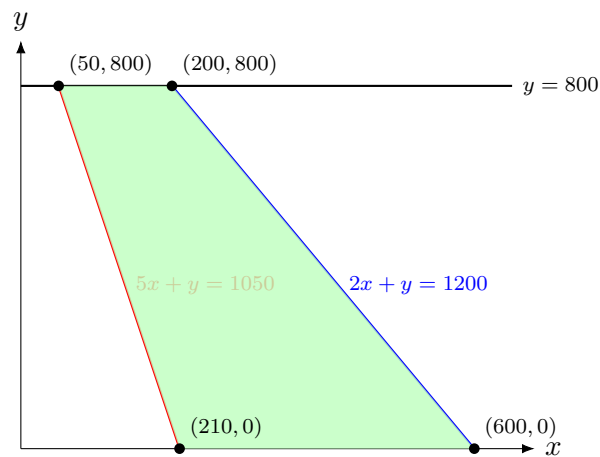
$$(50, 800), \quad (200, 800), \quad (600, 0), \quad (210, 0).$$

Evaluate $P = 5x + 2y$ at these points:

Corner point	$P = 5x + 2y$
$(50, 800)$	$250 + 1600 = 1850$
$(200, 800)$	$1000 + 1600 = 2600$
$(600, 0)$	3000
$(210, 0)$	1050

The maximum value is 3000 at $(600, 0)$.

$$P_{\max} = 3000 \text{ at } (x, y) = (600, 0).$$



Theory and Formula Sheet for This Paper

Determinants and Matrices

- Row/column factors can be taken outside a determinant.
- If $AB = BA = I$, then $B = A^{-1}$.
- If $AB = BA = kI$, then $A^{-1} = \frac{1}{k}B$.
- Cramer's rule: for $AX = B$, $x_i = \Delta_i/\Delta$, where $\Delta = |A| \neq 0$.
- Log rule: $\ln(ab) = \ln a + \ln b$, $\ln(a^r) = r \ln a$.
- $\frac{d}{dx}a^x = a^x \ln a$.
- $\int a^x dx = \frac{a^x}{\ln a} + C$ for $a > 0$, $a \neq 1$.
- Function increasing when $f'(x) > 0$ and decreasing when $f'(x) < 0$.

Sequences

- A.P.: $T_n = a + (n-1)d$.
- H.P.: reciprocals of H.P. terms form an A.P.
- Finite G.P.: $S_n = a \frac{1-r^n}{1-r}$ for $r \neq 1$.
- Sum of first n odd numbers: $1+3+\dots+(2n-1) = n^2$.

Complex Numbers

- If $z = x+iy$, then $|z| = \sqrt{x^2+y^2}$ and $\text{Im}(z) = y$.
- For real-coefficient quadratic equations, non-real complex roots occur in conjugate pairs.
- Cube roots of unity: $1, \omega, \omega^2$ with $1+\omega+\omega^2 = 0$ and $\omega^3 = 1$.

Calculus

- Quotient rule: $\left(\frac{u}{v}\right)' = \frac{vu' - uv'}{v^2}$.

Vectors and 3D Geometry

- Direction cosines of vector (l, m, n) are $\left(\frac{l}{\sqrt{l^2+m^2+n^2}}, \frac{m}{\sqrt{l^2+m^2+n^2}}, \frac{n}{\sqrt{l^2+m^2+n^2}}\right)$.
- A vector perpendicular to both $\vec{a}$ and $\vec{b}$ is $\vec{a} \times \vec{b}$.
- Unit vector in direction $\vec{v}$ is $\vec{v}/|\vec{v}|$.
- Scalar triple product: $(\vec{a} \times \vec{b}) \cdot \vec{c}$.
- Diagonals of a rhombus: if side vectors are $\vec{a}, \vec{b}$ and $|\vec{a}| = |\vec{b}|$, then $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 0$.

Inequalities and LPP

- For rational inequalities, bring to one side, find critical points, and test intervals.
- In linear programming, optimum occurs at a corner point of the feasible region.
- Evaluate the objective function at all feasible corner points and choose maximum/minimum.